



Required Capstone Project: Module 12

Due Sep 2 by 11:59pm**Points** None**Available** after Aug 19 at 12am

✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528589)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528591)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528592)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528593)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528594)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528595)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528596)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528597)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528598)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528599)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528600)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528601)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528602)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528603)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528604)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528605)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528606)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528608)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528610)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528611)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528612)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528613)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528614)
✓		(https://classroom.emeritus.org/courses/16839/modules/items/3528615)

✓	📄	(https://classroom.emeritus.org/courses/16839/modules/items/3528616)
✓	📄	(https://classroom.emeritus.org/courses/16839/modules/items/3528617)
✓	📄	(https://classroom.emeritus.org/courses/16839/modules/items/3528618)
✓	📄	(https://classroom.emeritus.org/courses/16839/modules/items/3528619)
✓	🔗	(https://classroom.emeritus.org/courses/16839/modules/items/3528620)
✓	💬	(https://classroom.emeritus.org/courses/16839/modules/items/3528621)
✓	💬	(https://classroom.emeritus.org/courses/16839/modules/items/3528622)
✓	📄	(https://classroom.emeritus.org/courses/16839/modules/items/3528624)
📍	✓	📄 (https://classroom.emeritus.org/courses/16839/modules/items/3528626)
✓	🔗	(https://classroom.emeritus.org/courses/16839/modules/items/3528628)
✓	💬	(https://classroom.emeritus.org/courses/16839/modules/items/3528629)
✓	📄	(https://classroom.emeritus.org/courses/16839/modules/items/3528630)

🎯 Learning Outcomes Addressed

- Understand how AI, GenAI and Agentic AI work and where they apply in business.
- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 5 – Deployment & Governance

Milestone 12: AI Governance & Control Framework

Estimated Effort: 2 Hours

Objective

As organisations deploy AI systems in real operational environments, they must carefully manage risks associated with automated decision-making.

AI systems can introduce challenges related to:

- bias and fairness in outputs
- hallucinations and misinformation
- data privacy and protection
- regulatory compliance
- operational accountability

Without proper governance, AI systems may produce unreliable outputs, create regulatory exposure, or damage organisational trust.

For this reason, organisations implement **AI governance frameworks** that ensure AI systems operate responsibly, transparently, and in alignment with organisational policies and regulatory expectations.

In this module, you will design a **Governance & Control Framework** that enables your AI solution to operate safely within an enterprise environment.

How This Connects to Earlier Modules

In earlier modules, you designed:

- the AI-enabled workflow
- agent architecture and automation logic
- human-in-the-loop governance
- enterprise deployment architecture

In this module, you focus on **risk management and governance**.

You will define how the organisation:

- monitors the behaviour of the AI system
- mitigates potential risks
- maintains accountability for AI-generated outcomes

Note: This is a required assignment and counts towards programme completion.

Task Overview

Create an **AI Governance & Control Framework** that outlines how risks associated with your AI solution will be identified, monitored, and managed.

Your framework should define the following governance mechanisms:

1. Data privacy controls
2. Hallucination mitigation strategies
3. Bias monitoring mechanisms
4. Audit trail and traceability
5. Incident response plan
6. Regulatory and compliance considerations

Your framework should demonstrate that the AI system can operate **responsibly and safely within an organisational environment**.

Step 1 — Identify Key AI Risks

Start by identifying the main **risks associated with your AI system**.

Risks may arise from:

- incorrect or misleading outputs
- misuse of sensitive organisational data
- unintended bias in automated decisions
- lack of transparency in AI-generated outcomes

Focus on the **most relevant risks for your specific AI workflow**.

Example (Illustrative)

Risk Type	Example Risk
Hallucination	AI generates incorrect information in automated responses
Data Privacy	Sensitive customer data exposed during processing
Bias	AI outputs favour certain customer groups
Operational Risk	AI agent performs an incorrect workflow action

Your goal is to demonstrate **awareness of potential AI risks**.

Step 2 — Define Data Privacy Controls

AI systems often interact with organisational data, which may include sensitive or personal information.

Your governance framework should explain how **data privacy is protected**.

Consider controls such as:

- data access restrictions
 - encryption of sensitive data
 - anonymisation or masking of personal information
 - limiting the data available to AI models
-

Example

Customer data used by the AI system is anonymised before processing, and access to the data environment is restricted to authorised systems and personnel.

Step 3 — Define Hallucination Mitigation Mechanisms

Generative AI systems may occasionally produce **incorrect or fabricated outputs**, often referred to as hallucinations.

Your framework should define mechanisms to reduce this risk.

Possible mitigation approaches include:

- grounding AI outputs in trusted data sources
 - retrieval systems that provide factual context
 - human approval checkpoints for critical outputs
 - automated validation checks
-

Example

AI-generated responses are validated against internal knowledge base documents before being sent to users.

Step 4 — Define Bias Monitoring Mechanisms

AI systems can unintentionally reflect biases present in training data or operational processes.

Your governance framework should include mechanisms to **monitor and address potential bias**.

Possible approaches include:

- regular output audits
 - fairness testing across user groups
 - human review for sensitive decisions
 - retraining models when bias is detected
-

Example

A quarterly review evaluates AI-generated outputs to ensure consistent treatment across customer segments.

Step 5 — Define Audit Trails and Traceability

Organisations must be able to **trace AI system behaviour** to maintain accountability.

An audit trail ensures that key actions and outputs can be reviewed when needed.

Your framework should define how the organisation records:

- AI-generated outputs
 - workflow actions triggered by AI
 - human overrides or approvals
 - system errors or anomalies
-

Example

All AI decisions and agent actions are logged in an internal monitoring system, enabling retrospective analysis of system behaviour.

Step 6 — Design an Incident Response Plan

Despite safeguards, AI systems may still experience failures or unexpected behaviour.

Your governance framework should define **how incidents are detected and managed**.

An incident response plan may include:

- automated error detection alerts
 - escalation to human operators
 - temporary suspension of automated workflows
 - investigation and remediation procedures
-

Example

If the AI system generates repeated incorrect outputs, the automation is paused and the issue is escalated to the operations team for review.

Step 7 — Consider Regulatory and Compliance Requirements

Many industries operate under **regulatory frameworks that govern the use of data and automated systems**.

Your governance design should consider relevant regulatory expectations, such as:

- data protection regulations
- consumer protection rules
- transparency requirements for automated decision-making

You do not need to conduct a full legal analysis, but you should demonstrate **awareness of compliance considerations**.

Example

The AI system complies with organisational data protection policies and ensures that sensitive customer data is processed securely.

Required Output Structure

Your **AI Governance & Control Framework** should include:

1. AI Risk Identification

Summary of key risks associated with your AI solution.

2. Governance Controls

Description of mechanisms addressing privacy, hallucinations, and bias.

3. Monitoring and Audit Processes

Explanation of how the organisation tracks AI behaviour and maintains accountability.

4. Incident Response Strategy

Outline of procedures for managing AI system failures or unexpected behaviour.

5. Regulatory Considerations

Brief reflection on regulatory or compliance considerations relevant to the AI system.

Practical Tips

Focus on Realistic Governance Mechanisms

Your governance framework should include **practical controls that organisations commonly implement**, rather than theoretical concepts.

Align Governance With Your Workflow Design

Your controls should address the **specific risks introduced by your AI workflow and automation logic**.

Balance Automation and Oversight

Effective AI governance ensures that AI systems can operate efficiently while maintaining **appropriate human oversight and accountability**.

Avoid These Common Mistakes

- ✗ Assuming AI systems always produce correct outputs
- ✗ Ignoring data privacy risks

✗ Designing governance frameworks without monitoring mechanisms

✗ Failing to define how incidents will be handled

Strong governance frameworks ensure that AI systems remain **trustworthy, transparent, and accountable**.

Submission

Before submitting your **AI Governance & Control Framework**, confirm that you have included:

- ✓ Identification of key AI risks
- ✓ Governance controls addressing privacy, hallucinations, and bias
- ✓ Monitoring and audit processes
- ✓ Incident response strategy
- ✓ Regulatory or compliance considerations

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 20

Criteria	Full Points	0 Points
AI Risk Identification (4 points)	The submission identifies the key AI risks associated with the proposed AI solution. (4 points)	The key AI risks associated with the AI solution are missing. (0 points)
Governance Controls (5 points)	The submission explains the governance controls for data privacy, hallucination mitigation, and bias monitoring required for the AI solution. (5 points)	The governance controls for data privacy, hallucination mitigation, and bias monitoring are missing. (0 points)
Monitoring & Audit Processes (4 points)	The submission explains the monitoring processes, audit trail, and traceability mechanisms required to monitor the AI solution. (4 points)	The monitoring processes, audit trail, and traceability mechanisms are missing. (0 points)
Incident Response Strategy (4 points)	The submission explains the incident response strategy for detecting, managing, and resolving AI system failures or unexpected behaviour. (4 points)	The incident response strategy is missing. (0 points)
Regulatory & Compliance Considerations (3 points)	The submission identifies the relevant regulatory or compliance considerations for the AI solution. (3 points)	The regulatory or compliance considerations are missing. (0 points)

Why this Step Matters

AI systems introduce **new operational and ethical challenges** that organisations must manage carefully.

Strong governance frameworks help organisations:

- maintain trust in AI systems
- protect sensitive data

- comply with regulatory expectations
- ensure accountability for automated decisions

Responsible AI governance is therefore a critical foundation for **scaling AI adoption across the enterprise.**

In the final module, you will bring together all elements of your capstone work into a **complete AI solution proposal and executive presentation**